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GTN (Glyceryl Trinitrate) Spray

Mechanism of action

- GTN is a vasodilator. It acts on vascular smooth muscle to cause venous and arterial vasodilation, with the predominant effect being on veins.
- The mechanism of action is not clear, but it appears that GTN results in the formation of nitric oxide which is a vasodilator. GTN causes:
  - A reduction in venous return (preload) to the heart. This reduces ventricular filling and cardiac output which reduces myocardial oxygen demand.
  - Arterial dilation which reduces peripheral resistance (afterload). This reduces the force the left ventricle must overcome to eject blood into the arteries which reduces myocardial oxygen demand.
  - Dilation of the coronary arteries which may increase coronary blood supply, though this is not usually clinically significant.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Myocardial ischaemia.
- ✓ Cardiogenic pulmonary oedema.
- ✓ Control of hypertension associated with autonomic dysreflexia.
- ✓ Control of hypertension (usually in conjunction with labetalol) prior to fibrinolytic treatment for STEMI.
- ✓ Control of hypertension (usually in conjunction with labetalol) during inter-hospital transfer for STEMI.
- ✓ STEMI

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Contraindications

- ✕ Known severe allergy.
- ✕ Systolic BP less than 110 mmHg.
- ✕ Heart rate less than 40/minute.
- ✕ Heart rate greater than 150/minute.
- ✕ Ventricular tachycardia.

Cautions

- ⚠ STEMI, particularly STEMI involving the right ventricle. GTN may cause a significant fall in cardiac output and if there are signs of low cardiac output GTN should be withheld.
- ⚠ The patient is frail.
- ⚠ Signs of shock. The patient's reduced cardiac output may fall further with GTN administration.
- ⚠ Dysrhythmia. This may cause a reduced cardiac output which may fall further with GTN administration.
- ⚠ Has taken a phosphodiesterase inhibitor in the last 24 hours. See common interactions.
- ⚠ Known aortic or mitral stenosis. With aortic or mitral stenosis, cardiac output may be reduced as a result of the narrowed valve and a fall in preload may cause a further fall in cardiac output.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of a pregnant or breastfeeding patient requiring GTN is very low, but GTN should be administered if indicated.

Dosage

- Myocardial ischaemia: 0.4 mg every five minutes. Consider increasing the dosing interval to ten minutes if a caution is present.
- STEMI: 0.4 mg with caution. Withhold if signs of poor perfusion are present.
- Cardiogenic pulmonary oedema: 0.8 mg every five minutes. Consider increasing the dosing interval to ten minutes if a caution is present. Increase the dose and frequency if the patient is not improving.

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- Control of hypertension: 0.4-0.8 mg every five minutes as required.

Administration

- Spray under the tongue. If this cannot be achieved it is acceptable to spray into the mouth.
- If GTN is administered in the presence of a caution:
  - The patient should be lying flat.
  - IV access should have been obtained whenever possible.
  - The dosing interval should be increased to ten minutes.
  - Be ready to administer 0.9% sodium chloride IV if there is a significant fall in cardiac output or blood pressure.

Common adverse effects

- Hypotension.
- Flushing.
- Headache.
- Tachycardia.
- Feeling light-headed.

Usual onset of effect

- 1-2 minutes.

Usual duration of effect

- 15-30 minutes.

Usual preparation

- Metered dose bottle delivering 0.4 mg doses.

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Pharmacokinetics

- GTN is rapidly absorbed from the oral mucosa and reaches the vascular system without passing through the liver.
- GTN is predominantly metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The effects may be increased if the patient is taking an antihypertensive medicine.
- Severe and/or prolonged hypotension may occur if a phosphodiesterase inhibitor has been taken within the last 24 hours:
  - Phosphodiesterase inhibitors can be used in the treatment of erectile dysfunction and pulmonary hypertension (particularly sildenafil).
  - These medicines are long-acting vasodilators and the administration of GTN may cause further vasodilation.

Additional information

- GTN must be used with caution in the presence of STEMI because the risks may outweigh the benefits:
  - GTN may cause a significant fall in cardiac output.
  - GTN has a role in treating symptomatic myocardial ischaemia, but does not usually have a significant role in treating STEMI.
- Particular caution must be used in the presence of STEMI involving the right ventricle and personnel must have a low threshold for withholding GTN:
  - STEMI involving the right ventricle can result in a significant reduction in right ventricular contractility.
  - When the right ventricle is significantly impaired, it may provide little in the way of contribution to cardiac output and blood may be passively flowing down a pressure gradient between the inferior vena cava (IVC), the superior vena cava (SVC) and the left atrium.
  - This may result in the preload (filling) of the left side of the heart being dependent on the venous pressures within the IVC and SVC.
  - GTN can result in a significant fall in venous pressure (and thus a fall in preload) which may cause a significant fall in cardiac output.
- GTN and VT:

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- GTN is not administered for chest pain associated with VT because the risk of precipitating cardiac arrest outweighs the benefits.
- During VT the atria and ventricles are contracting independently of each other (also called atrioventricular dissociation). This causes a reduction in ventricular preload (or filling) through loss of the 'atrial kick', which in combination with abnormal ventricular contraction may cause a substantial fall in cardiac output.
- GTN reduces venous return to the heart and may further reduce cardiac output with the risk of precipitating severe hypotension or cardiac arrest.

More Information

Version history

16 June 2026 1.1.0.1
Update: Dosage updated.

11 September 2023 1.0.4
Update: Multiple changes.

22 November 2022 1.0.1
Update: Cautions.